MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
3000 PROCESS PLANT LIGHTING
DESIGN OF FOUNDATION FOR
HIGH MAST TOWER

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of 30m High mast Pole Foundation in Process Area.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of 30m High mast Pole Foundation only.

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4. DESIGN METHODOLOGY & ASSUMPTIONS

Pole is 30m High mast pole consisting of almost 12 Flood Light placed above. These Light Poles are located all around the Process Plat. The Foundations have been designed based on the Specifications & Weights provided by Vendor which are attached in the Appendix Part of this Report.

4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008-A02.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

Fy = 414 MPa (60 Ksi) for all reinforcing steel.

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5. DESIGN LOADS

5.1. Dead Load

5.1.1. Pole Self Weight

The Weight provided by the Vendor = 1938 Kgs = **19.4 kN**

5.2. Wind Load

The Maximum Horizontal Shear Provided by the Vendor =1511 Kgs = **15.2 kN**

The Maximum Moment provided by the Vendor = 33217 Kg-m = **332 kN-m**

6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + W

D + 0.75W

0.6D + W

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.2D

1.2D + 0.8W

1.2D + 1.6W

0.9D + 1.6W

Where:

D = Dead Load

W = Wind Load

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7. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 2.8m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008-A01 for Details)

Refer ATTACHMENT 2 for Foundation Design calculations

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REFERENCES

1.)	BUILDING PLANS	MD-216-6000-EG-AR-GAD-3092
2.)	OVERALL LAYOUT PLAN	MD-216-6000-EG-GAD-1535
3.)	ILLUMINATION & SMALL POWER OUTDOOR LAYOUT	MD-216-6000-EG-EL-LAD-1021
4.)	GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
5.)	STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

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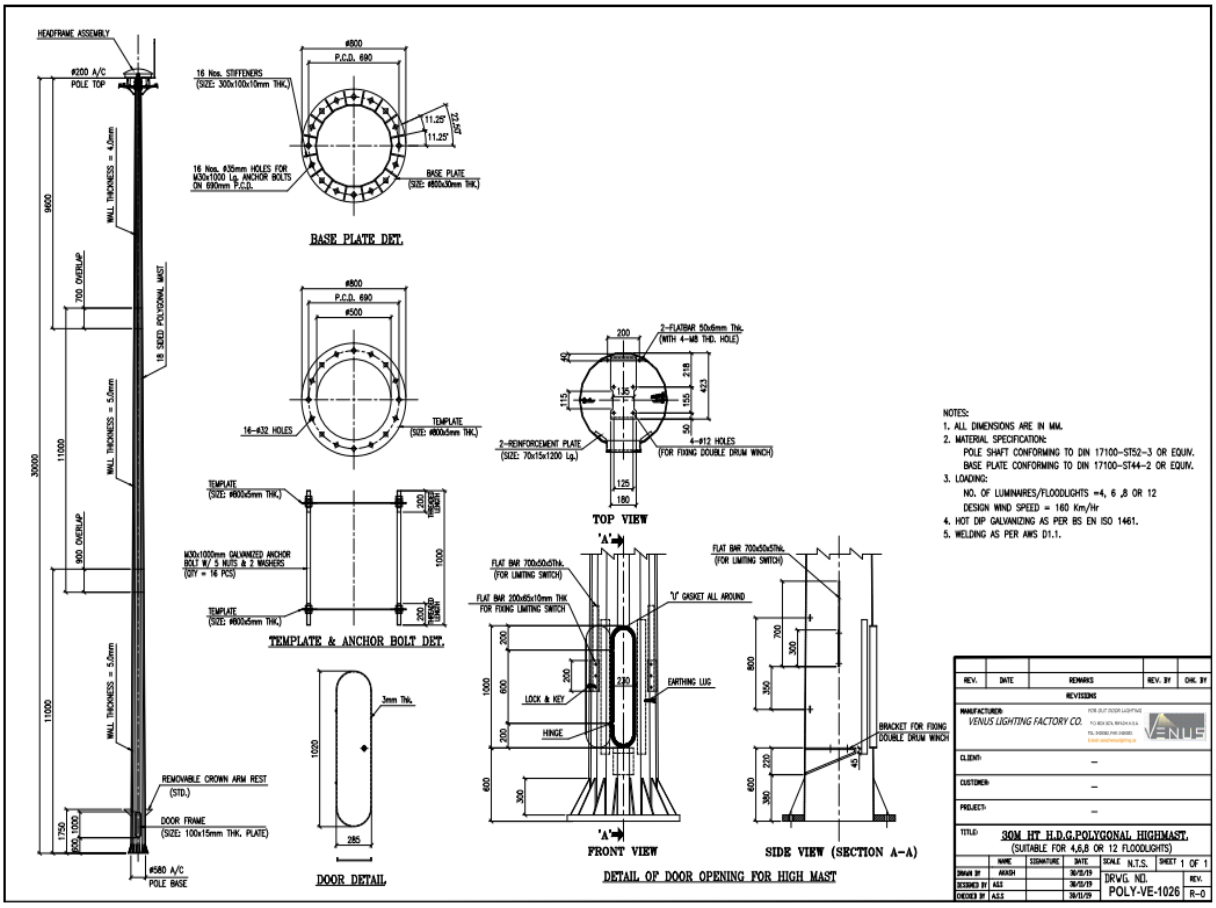
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ATTACHMENT 1 VENDOR DRAWINGS & DETAILS

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Ground line reactions (from computer printout) HR-10162

Maximum bending moment, M	=	33217	kg-m
Maximum axial force, V	=	1938	kg
Maximum shear force, S	=	1511	kg

ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

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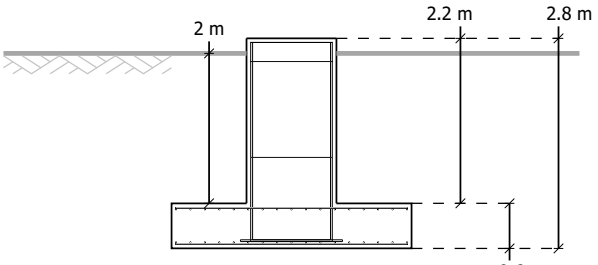
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Isolated Footing Design(ACI 318-11)

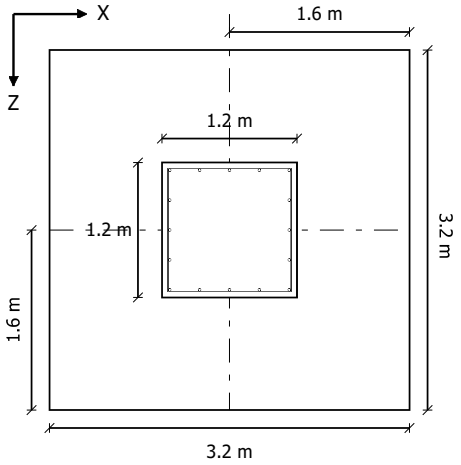
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
1	1	3.200m	3.200m	0.600m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
1	17 - ϕ20	17 - ϕ20	17 - ϕ20	17 - ϕ20	16 - #25	#6 @ 50 mm

Isolated Footing 1



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 600.000mm
Footing Length - X (Fl) : 3200.000mm
Footing Width - Z (Fw) : 3200.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.300m

Column Width - Z (B_{col}) : 0.300m

Pedestal

Include Pedestal? Yes
Pedestal Shape : Rectangular
Pedestal Height (Ph) : 2.200m
Pedestal Length - X (Pl) : 1.200m
Pedestal Width - Z (Pw) : 1.200m

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 415.000N/mm2
Minimum Bar Size : #20
Maximum Bar Size : #20
Top Footing Minimum Bar Size : #20
Top Footing Maximum Bar Size : #20
Pedestal Minimum Bar Size : #25
Pedestal Maximum Bar Size : #25
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 200.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 2000.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Always calculate based on self weight
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 3.200m

Initial Width (W_o) = 3.200m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
101	1.000 x DL	1.00	1.00	1.00
103	1.000 x DL+1.000 x LL	1.00	1.00	1.00
107	1.000 x DL+0.750 x LL	1.00	1.00	1.00
110	1.000 x DL+0.600 x WLX	1.00	1.00	1.00
114	1.000 x DL+0.600 x WLZ	1.00	1.00	1.00
122	1.000 x DL+0.750 x LL+0.450 x WLX	1.00	1.00	1.00
128	1.000 x DL+0.750 x LL+0.450 x WLZ	1.00	1.00	1.00
142	0.600 x DL+0.600 x WLX	1.00	1.00	1.00
146	0.600 x DL+0.600 x WLZ	1.00	1.00	1.00
150	0.600 x DL	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
201	1.400 x DL	1.00	1.00	1.00
202	1.200 x DL+1.600 x LL	1.00	1.00	1.00
204	1.200 x DL+1.000 x LL	1.00	1.00	1.00
206	1.200 x DL+0.500 x WLX	1.00	1.00	1.00
208	1.200 x DL+0.500 x WLZ	1.00	1.00	1.00
214	1.200 x DL+1.000 x LL+1.000 x WLX	1.00	1.00	1.00
218	1.200 x DL+1.000 x LL+1.000 x WLZ	1.00	1.00	1.00
226	0.900 x DL+1.000 x WLX	1.00	1.00	1.00
228	0.900 x DL+1.000 x WLZ	1.00	1.00	1.00
230	0.900 x DL	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
101	20.000	0.000	0.000	0.000	0.000
103	20.000	0.000	0.000	0.000	0.000
107	20.000	0.000	0.000	0.000	0.000
110	20.000	9.000	0.000	0.000	-199.200
114	20.000	0.000	0.000	0.000	0.000
122	20.000	6.750	0.000	0.000	-149.400
128	20.000	0.000	0.000	0.000	0.000
142	12.000	9.000	0.000	0.000	-199.200
146	12.000	0.000	0.000	0.000	0.000
150	12.000	0.000	0.000	0.000	0.000

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
201	28.000	0.000	0.000	0.000	0.000
202	24.000	0.000	0.000	0.000	0.000
204	24.000	0.000	0.000	0.000	0.000
206	24.000	7.500	0.000	0.000	-166.000
208	24.000	0.000	0.000	0.000	0.000
214	24.000	15.000	0.000	0.000	-332.000
218	24.000	0.000	0.000	0.000	0.000
226	18.000	15.000	0.000	0.000	-332.000
228	18.000	0.000	0.000	0.000	0.000
230	18.000	0.000	0.000	0.000	0.000

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 10.240 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 2.472 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) =	3.200	m	Governing Load Case :	# 101
Width (W_2) =	3.200	m	Governing Load Case :	# 101
Depth (D_2) =	0.600	m	Governing Load Case :	# 214

Depth is governed by Ultimate Load Case

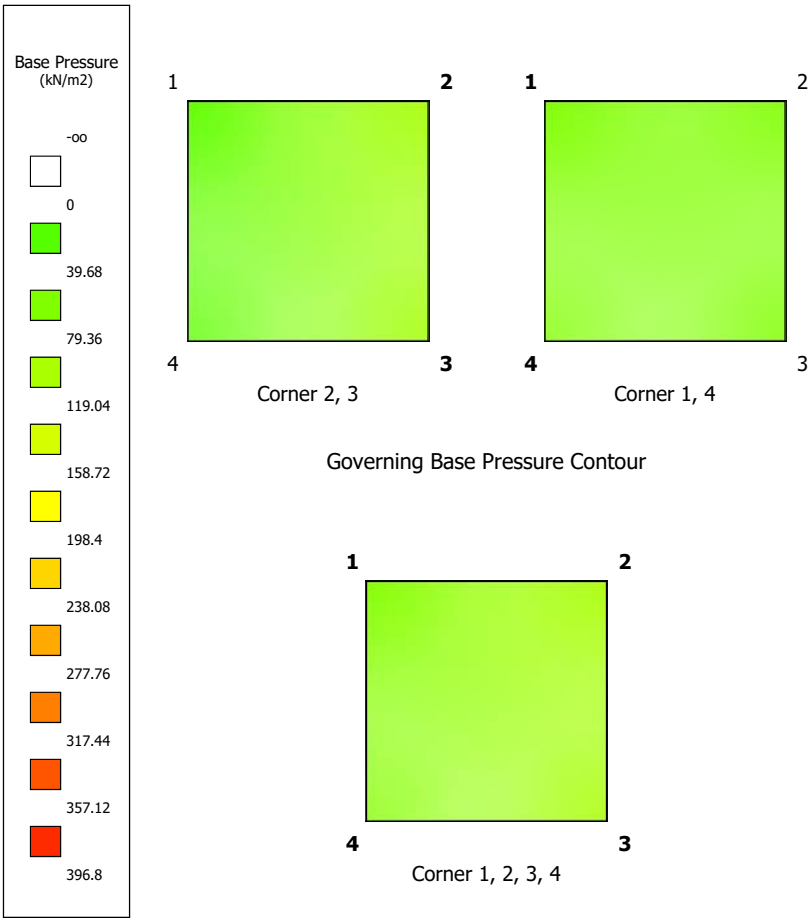
(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) =	10.240	m^2
Final Pedestal Height =	2.200	m
Final Soil Height =	2.000	m
Footing Self Weight =	223.480	kN
Gross Soil Bearing Capacity =	396.80	kN/m2
Soil Weight On Top Of Footing =	316.789	kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



Governing Base Pressure Contour

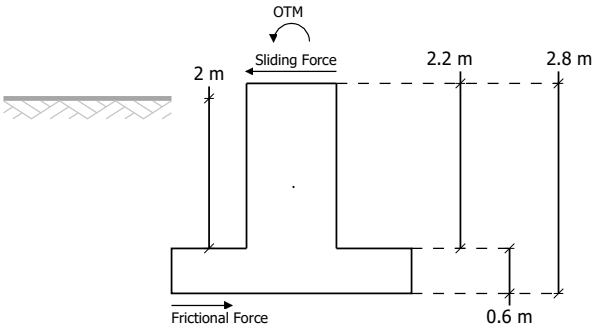
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
101	**54.7138**	54.7138	54.7138	54.7138	0.000
110	13.6242	**95.8033**	95.8033	13.6242	0.000
110	13.6242	95.8033	**95.8033**	13.6242	0.000
101	54.7138	54.7138	54.7138	**54.7138**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
101	**54.7138**	54.7138	54.7138	54.7138
110	13.6242	**95.8033**	95.8033	13.6242
110	13.6242	95.8033	**95.8033**	13.6242
101	54.7138	54.7138	54.7138	**54.7138**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
101	N/A	N/A	N/A	1.500	N/A	N/A	1.500
103	N/A	N/A	N/A	1.500	N/A	N/A	1.500
107	N/A	N/A	N/A	1.500	N/A	N/A	1.500
110	32.994	N/A	46.660	1.500	N/A	4.180	1.500
114	N/A	N/A	N/A	1.500	N/A	N/A	1.500
122	43.991	N/A	62.213	1.500	N/A	5.573	1.500
128	N/A	N/A	N/A	1.500	N/A	N/A	1.500
142	32.523	N/A	45.994	1.500	N/A	4.122	1.500
146	N/A	N/A	N/A	1.500	N/A	N/A	1.500
150	N/A	N/A	N/A	1.500	N/A	N/A	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 142

Governing Disturbing Force :	9.000kN
Governing Restoring Force :	292.703kN
Minimum Sliding Ratio for the Critical Load Case :	32.523
Critical Load Case for Overturning about X-Direction :	N/A
Governing Overturning Moment :	N/A
Governing Resisting Moment :	N/A
Minimum Overturning Ratio for the Critical Load Case :	N/A

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction :	N/A
Governing Disturbing Force :	N/A
Governing Restoring Force :	N/A
Minimum Sliding Ratio for the Critical Load Case :	N/A
Critical Load Case for Overturning about Z-Direction :	142
Governing Overturning Moment :	-224.400kNm
Governing Resisting Moment :	925.084kNm
Minimum Overturning Ratio for the Critical Load Case :	4.122

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

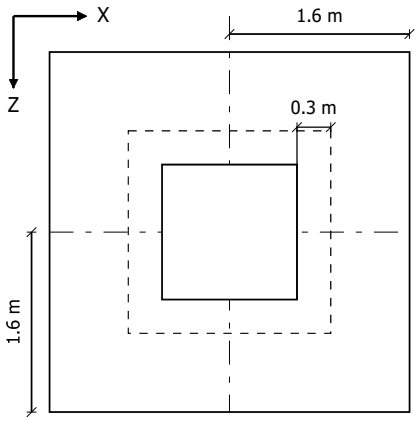
Critical Load Case for Sliding along Resultant Direction :	142
Governing Disturbing Force :	9.000kN
Governing Restoring Force :	413.944kN
Minimum Sliding Ratio for the Critical Load Case :	45.994

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D =	0.600m	
Calculated Effective Depth, d_{eff} =	D - C_{cover} - 0.5 * d_b =	0.540m
For rectangular column, β_c =	B_{col} / D_{col} =	1.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 453.257kN, Load Case # 214

From ACI Cl.11.11.2, b_o for column=	$2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ =	6.960m
Equation 11-31, V_{c1} =	$0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$	11077.656kN
Equation 11-32, V_{c2} =		9422.374kN

$$0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2 \right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff} \right)$$

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 7385.104kN

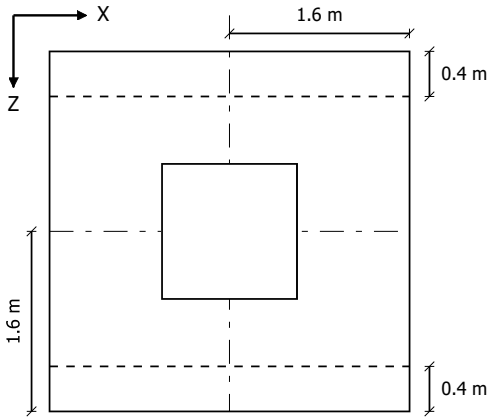
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 5538.828kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 1697.725kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.460m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

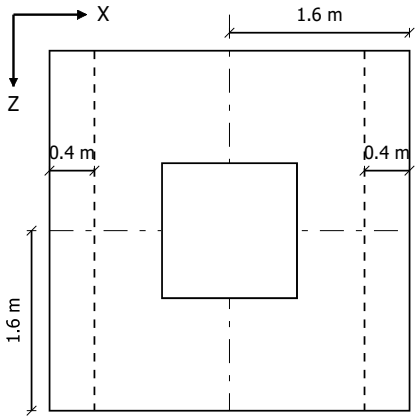
From above calculations, 0.75 X V_c = 1273.294 kN

Critical load case for V_{ux} is # 214 $V_{ux} = V_{ux}|_{x=D_x}$ = 82.044 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

1697.725 kN

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ =

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.460 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

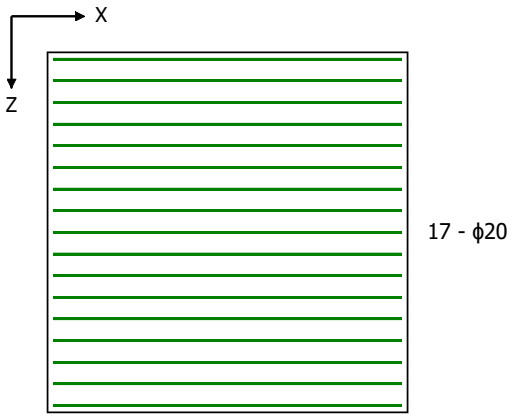
From above calculations, 0.75 X V_c = 1273.294 kN

Critical load case for V_{uz} is # 214 $V_{uz} = V_{uz}|_{z=D_z}$ = 160.757 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 226

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.540 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ = 0.03374

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02530

From ACI Cl. 7.12.2, ρ_{min} = 0.00176

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 13.950

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x = $0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ = 2.200 m

Ultimate moment, $M_u|_{z=D_z}$ = 189.293 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 210.326 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{(F_y \times W \times d_{eff}^2)}}\right]$ = 0.00055

(Based on gross depth) ρ x d_{eff} / Depth = 0.00049

Since	$\rho = \rho_{min}$	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	3378.073	mm2

Selected bar Size = #20

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 1st

Max spacing for Cracking Consideration = 254.793mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#20 @ 200mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ =	=0.845	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) \cdot - C_{cover}$ =	0.950	m
Try bar size # 20	Area of one bar =	314.161	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	17	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	5340.731	mm2
d_{eff} =	D - C_{cover} - 0.5 X (dia. of one bar) =	0.540	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00309	

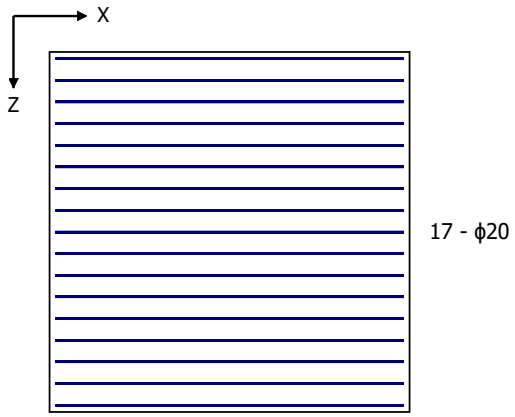
From ACI Cl.7.6.1, minimum req'd clear distance between bars

C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.581

Check to see if width is sufficient to accomodate bars

Design For Top Reinforcement Parallel to X Axis



Top reinforcement is calculated based on self weight of footing and soil

Calculate the flexural reinforcement for M_z. Find the area of steel required

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.540	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530	
From ACI Cl.7.12.2, ρ_{min} =		0.00000	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	13.950	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.000	m
Ultimate moment,	$M_u\big\vert_{x=D_x}$ =	80.636	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	89.595	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.000232	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.000209	
Since	$\rho_{min} < \rho < \rho_{max}$	OK	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	400.455	mm2
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	5340.686	mm2

Provided Steel Area / Required Steel Area = 13.337

Selected bar Size = #20

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 2nd

Max spacing for Cracking Consideration = 254.793mm

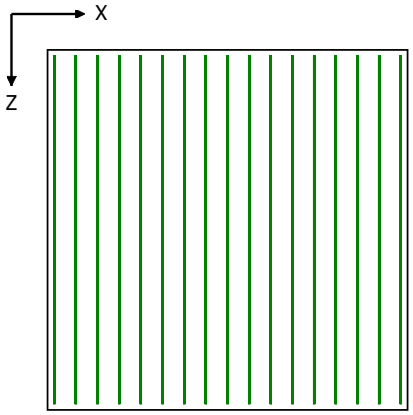
Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#20 @ 200mm o.c.

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



17 - ϕ20

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 214

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.520	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530	
From ACI Cl.7.12.2, ρ_{min} =		0.00178	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	13.950	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z =	$0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ =	2.200	m
Ultimate moment,	$M_u\big\vert_{x=D_x}$ =	93.059	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	103.399	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00029	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00025	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	3411.109	mm2

Selected Bar Size = #20

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

According to ACI 318 Clause No- 10.6.4 3rd

Max spacing for Cracking Consideration = 254.793mm

Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

#20 @ 200mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.845	m
Available development length for bars, D_L =	$0.5 \times (L - D_{col}) \cdot - C_{cover}$ =	0.950	m
Try bar size # 20	Area of one bar =	314.161	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	17	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	5340.731	mm2
d_{eff} =	D - C_{cover} - 1.5 X (dia. of one bar) =	0.520	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{(d_{eff} \times W)}$ =	0.00321	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

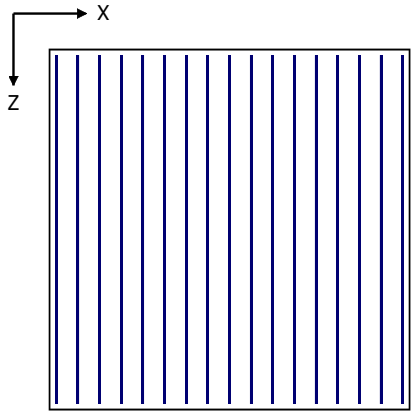
Provided Steel Area / Required Steel Area = 1.566

Check to see if width is sufficient to accomodate bars

Bending moment for uplift cases will be calculated based solely on selfweight, soil depth and surcharge loading.

As the footing size has already been determined based on all servicebility load cases, and design moment calculation is based on selfweight, soil depth and surcharge only, top reinforcement value for all pure uplift load cases will be the same.

Design For Top Reinforcement Parallel to Z Axis



17 - ϕ20

Top reinforcement is calculated based on self weight of footing and soil

Calculate the flexural reinforcement for M_x. Find the area of steel required

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.520	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03374	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02530	
From ACI Cl. 7.12.2, ρ_{min} =		0.00000	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{(0.85 \times F_c')}$ =	13.950	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.000	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	80.636	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	89.595	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00025	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00022	
Since	$\rho_{min} < \rho < \rho_{max}$	OK	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	415.910	mm2
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	5340.686	mm2

Provided Steel Area / Required Steel Area = 12.841

Selected bar Size = #20

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

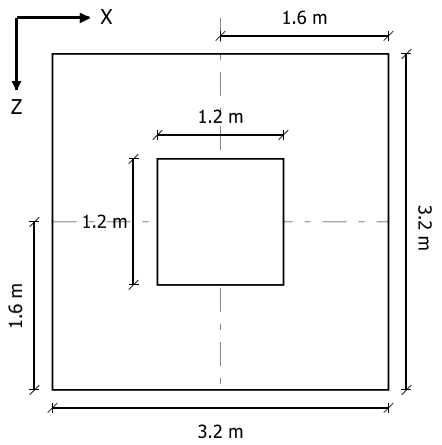
According to ACI 318 Clause No- 10.6.4 4th

Max spacing for Cracking Consideration = 254.793mm

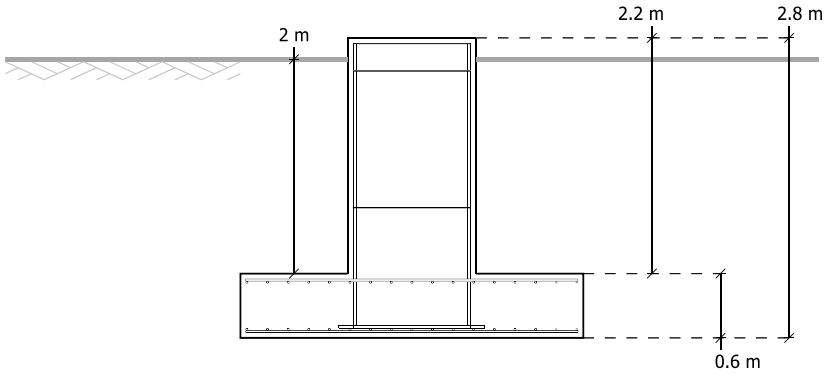
Safe for Cracking Aspect.

Based on spacing reinforcement increment; provided reinforcement is

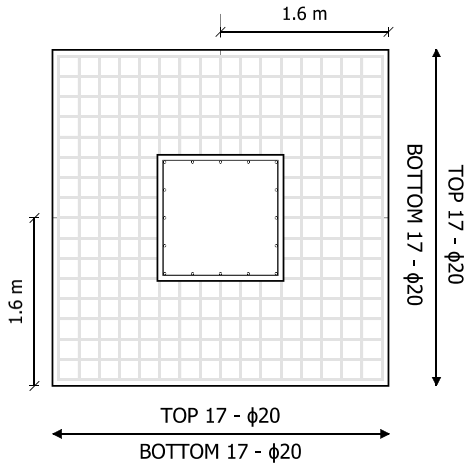
#20 @ 200mm o.c.



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	3.2 m	3.2 m	0.6 m	17 - ϕ20	17 - ϕ20	17 - ϕ20	17 - ϕ20
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	1.2 m	1.2 m	2.2 m	#25 - 16		#6 @ 50 mm	

Print Calculation Sheet